

Vacuum Tube Numbering Schemes

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Frequently, questions are asked about Vacuum Tube numbering schemes, so here is a few bits and pieces to start the ball rolling. This is by no means exhaustive, so if you feel you can add to this then please contact via the [forum](#) and I'll get the information included, with appropriate credits.

This page is lengthy, so you may wish to print it out.

[CV numbers](#) | [European scheme \(Mullard/Philips/General\)](#) | [Mazda power rectifiers](#) | [6As \(RETMA\) system](#)
CV numbers | European | European (old) | Mazda tubes | Mazda rectifiers | USA (RETMA)

Format is CV followed by up to 5 digits, e.g. CV4004. Used by the British Military to codify vacuum tubes, gas tubes, and latterly, some semiconductor devices. CV numbers can be directly mapped onto NATO numbers, as 5960-XX-000-YYYY where XX is the country code, and YYYY is the CV number. For example, 5960-99-000-4004 = CV4004 (12AX7WA), with 99 indicating country of origin as UK. **First letter: Filament**A4VB0.18A (series)**C**0.2A (series)**D**<= 1.4V (series/parallel)**E**6.3V (series/parallel)**F**12.6V**G**5V (parallel)**H**0.15A (series)**K**2V**L**0.45A (series)**P**0.3A (series)**U**0.1A (series)**V**0.05A (series)**X**0.6A (series)**Y**0.45A (series) **Second and subsequent letters:**
construction**A**Diode (excluding rectifier)**B**Double diode**C**Triode (signal, not power)**D**Power output triode**E**Tetrode (signal, not power)**F**Pentode (signal, not power)**L**Power output tetrode or pentode**H**Hexode or heptode (of the hexode type)**K**Octode or heptode (of the octode type)**M**Tuning indicator**Q**Nonode.**Y**Half wave rectifier**Z**Full wave rectifier **First digit: Basing****1**Miscellaneous**2**Miniature 10 pin.**3**International octal**4**8-pin miniature (B8A)**5**Magnoval (B9D)**8**Noval (B9A)**9**Miniature (B7G) **Remaining digits: sequence number***Note that signal pentodes and tetrodes which end in even numbers are sharp cutoff tubes. Those ending in odd numbers, are remote cutoff tubes. **Examples****EF86**: 6.3V filament, signal pentode, Noval base, sharp cutoff.**GZ34**: 5V filament, full wave rectifier, International Octal base.**PCL82**: 0.3A series string filament, signal triode + power pentode, Noval base.**E88CC**: Doesn't fit! This is actually a special quality version of the ECC88, and swapping of the second and third field was commonplace to denote "SQ" tubes. Many thanks to **Frank Philipse** for providing the following information. He kindly translated the Dutch text to English from: Gegevens en schakelingen van moderne ontvang-en versterkerbuizen. (1940 Philips).

Frank says "Initially type coding of Philips tubes was done in the following way: The type number consisted of a capital letter, followed by 3 or 4 digits. The capital letter indicated the heater current as follows:"

First letter: Heater current**A**0.06 to 0.10**B**0.10 to 0.20**C**0.20 to 0.40**D**0.40 to 0.70**E**0.70 to 1.25**F**1.25A and higher **Subsequent digits: Heater voltage, elements, amplification factor**The first digit, or in case of a four digit number the first two digits, indicated the heater voltage. For triodes, the last two digits indicated the amplification factor in it's working point. For tubes with more grids the last two digits had the following meaning:

41, 51, etc. were tetrodes with space charge grid. (dual grid tubes) 42, 52, etc. were H.F. screen grid tubes. (tetrode) 43, 53, etc. were output pentodes. 44, 54, etc. were binodes. (this was a combined diode-triode or diode-tetrode) 45, 55, etc. were H.F. tetrodes-selectodes (tetrode with variable gain) 46, 56, etc. were H.F. pentodes. 47, 57, etc. were H.F. pentodes-selectodes. (pentode with variable gain) 48, 58, etc. were hexode frequency changers. 49, 59, etc. were hexodes-selectodes. (hexode with variable gain)

An E499 for example is a triode. The E means that the heater current lies between 0.4 and 1.25 A (1.0 actually). The heater voltage is 4 Volts and the amplification factor is 99. An E446 is a H.F. pentode with a heater current between 0.4 and 1.25 A (1.1A actually) and a heater voltage of 4 Volts. Eventually, because of new developed tubes, this type numbering system was no longer sufficient and a new system was developed. Since 1934 all new tubes were coded according to the new system which is still in use for most European tubes. (e.g. ABC1, EABC80, AD1, EL34)

First number: Filament 11.4V (parallel or series) 66.3V (parallel or series) 100.1A (series) 200.2A (series) 300.3A (series) **Letters: construction** C Frequency changer with special oscillator section. D Signal diode(s). F Voltage amplifier tetrode or pentode. FD Voltage amplifier tetrode or pentode with diodes. FL Voltage amplifier tetrode or pentode with voltage amplifier triode. K Small gas triode or tetrode. L Voltage amplifier triode or double triode, including oscillator triode. LD Voltage amplifier triode with diode(s). M Tuning indicator P Power amplifier tetrode or pentode PL Power amplifier tetrode or pentode with voltage amplifier triode. **Final number: sequence number**

Examples: 6F22: 6.3V filament, Voltage amplifier tetrode or pentode. (6F22 = EF86). 6P15: 6.3V filament, Power amplifier tetrode or pentode. (6P15 = EL84 / 6BQ5). 30PL12: 0.3A series string filament, Power amplifier tetrode or pentode with voltage amplifier triode. (30PL12 = PCL82 / 16A8). **Letters indicate rectifier type:** U High vacuum half-wave. UU High vacuum full-wave. **Final number: sequence number** **First number: filament voltage** 0 Cold cathode (e.g. voltage regulator tube). 10.1 - 2.0 22.1 - 2.9 33.0 - 3.9 n n.0 - n.9 **Letter or letters: sequence code** One or two letters to indicate the sequence code. When the single character codes were used up, double character codes were introduced. Note that U - Z generally (but not always) as the only character, or last character, are used for rectifiers **Final number: Element count** The number of elements in the tube, the filament counting as 1. For example, the 6SN7 has two cathodes, two grids, two plates, and a filament. **Additional letters:** A Controlled heater warmup time, although can also be used to denote increased ratings or performance over and above the non-A model. B Improved ratings/performance. C Ditto. G Glass bulb. GT Glass tubular. W Ruggedised version. X Low loss ceramic base. Y Low loss phenolic base. **Examples:** 6SN7GT: Filament rating is between 6.0 - 6.9V, SN=sequence code, 7 elements, glass tubular envelope. 6V6GT: Filament rating is between 6.0 - 6.9V, V=sequence code, 6 elements, glass tubular envelope, low loss phenolic base. 12AY7: Filament rating is between 12.0 - 12.9V, AY=sequence code, 7 elements.